Project Design Phase-II Phase: Solution Requirements (Functional & Non-functional)

Date	**7 July 2026**
Team ID	
Project Name	**Nutrition Assistant**
Maximum Marks	**4 Marks**

1. Functional Requirements

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	User Registration	Registration using Name, Email, and Password Input validation for registration fields Secure password storage
FR-2	User Login & Authentication	Login using Email and Password
FR-3	Food Search	Search food items by name Display calories, protein, carbohydrates, fats, and other nutrition info Handle invalid or unavailable food searches
FR-4	Meal Management	Add food items to meal tracker View meal history Delete meals from tracker
FR-5	Dashboard	Display total calories consumed Display nutrition summary Show daily meal records
FR-6	User Profile	View user profile information Update basic profile details (optional)
FR-7	Data Management	Store user and meal information in MongoDB Retrieve user-specific nutrition records
FR-8	Error Handling	Display validation messages Handle API and server errors gracefully

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2. Non-functional Requirements

FR No.	Non-Functional Requirement	Description
NFR-1	Usability	The application should provide a simple, intuitive, and user-friendly interface that allows users to navigate easily without prior technical knowledge.
NFR-2	Security	User authentication must be implemented using JWT. Passwords should be securely encrypted, and user data must be protected from unauthorized access.
NFR-3	Reliability	The system should consistently provide accurate nutrition information and maintain data integrity with minimal failures.
NFR-4	Performance	Search results, dashboard loading, and meal operations should respond quickly with minimal delay under normal usage conditions.
NFR-5	Availability	The application should remain available whenever users need it, ensuring continuous access to nutrition tracking services.
NFR-6	Scalability	The architecture should support future growth by accommodating more users, larger datasets, and additional features such as AI recommendations and mobile integration.
NFR-7	Maintainability	The application should follow a modular MERN architecture, making it easier to update, debug, and maintain.
NFR-8	Compatibility	The application should function correctly on modern web browsers such as Google Chrome, Microsoft Edge, and Mozilla Firefox, and support responsive layouts for different screen sizes.